

Proof Compendium — Consumer Theory Block (L1–L4)

Advanced Microeconomics 1 · every result proved in the block, with statements

How to read this. The block builds two routes to the same matrix. The *choice-based* route (L1: R1–R5) assumes only consistency of observed choices; the *preference-based* route (L2: R6–R12) assumes a rational household and buys much more. The bridges (L3: R13–R17) are four derivative identities, and R15 is the meeting point: the choice-built S equals the preference-built $D_p h$. Recovery (L4: R18–R19, sketches) closes the circle. Statements are as presented in lecture (following MWG and the course notes); proofs are the lecture-appendix proofs, reformatted for the page. Results the course states but does not prove are listed at the end.

Lecture 1 — the choice-based route

Throughout: $x(p, w)$ is Walrasian demand (single-valued where differentiated), $S(p, w) = D_p x + D_w x x^\top$ the Slutsky matrix built from choices. A price change $p \rightarrow p'$ is (*Slutsky*) *compensated* if $w' = p' \cdot x(p, w)$.

R1 Proposition (MWG 1.B.2). Utility representation $\Rightarrow$ rationality

Statement. If a utility function $u : X \rightarrow \mathbb{R}$ represents $\succsim$ (that is, $x \succsim y \iff u(x) \geq u(y)$), then $\succsim$ is rational (complete and transitive).

Proof. *Completeness.* Take any $x, y \in X$. Then $u(x)$ and $u(y)$ are real numbers, and $\geq$ on $\mathbb{R}$ is complete: $u(x) \geq u(y)$ or $u(y) \geq u(x)$. Reading the representation right-to-left, $x \succsim y$ or $y \succsim x$. *Transitivity.* Suppose $x \succsim y$ and $y \succsim z$. By the representation, $u(x) \geq u(y)$ and $u(y) \geq u(z)$; transitivity of $\geq$ on $\mathbb{R}$ gives $u(x) \geq u(z)$; right-to-left again, $x \succsim z$. ■

$\succsim$ inherits the properties of $\geq$ on $\mathbb{R}$ through the representation. (The converse needs continuity: see Debreu, in the deferred list.)

R2 Proposition. Basket (Slutsky) compensation weakly over-compensates — the first policy result

Statement. Let $x = x(p, w)$ and compensate the household for $p \rightarrow p'$ so that its old bundle stays affordable: $w' = p' \cdot x$. Then the household at (p', w') is, by revealed preference, at least as well off as at (p, w) : full basket indexation is weakly more than the cost of keeping the old standard of living in reach.

Proof. The old bundle x lies in the new budget set by construction. If the household chooses $x' = x$, it is exactly as well off. If it chooses $x' \neq x$, it chose x' while x was affordable: x' is directly revealed at

least as good as x , and the old standard of living remains within reach. Either way, the compensation was (weakly) more than needed — because people substitute. ■

This is the substitution bias of fixed-basket indices, in three lines; the only assumption used is that choice reveals preference. Quantified at R12 and R17's expansion (L3): the over-payment is $-\frac{1}{2}\Delta p^\top S \Delta p$ to second order.

R3 Theorem (MWG 2.F.1). WARP and the compensated law of demand

Statement. Suppose $x(p, w)$ is homogeneous of degree zero and satisfies Walras' law. Then x satisfies WARP $\iff$ for every compensated price change,

$$(p' - p) \cdot [x(p', w') - x(p, w)] \leq 0, \quad \text{strict if } x(p', w') \neq x(p, w).$$

Proof. (*WARP $\Rightarrow$ law.*) If $x(p', w') = x(p, w)$ the claim is trivial. So take $x' \equiv x(p', w') \neq x \equiv x(p, w)$ and expand:

$$(p' - p) \cdot (x' - x) = p' \cdot (x' - x) - p \cdot (x' - x).$$

First term is zero: $p' \cdot x' = w'$ (Walras' law) and $p' \cdot x = w'$ (definition of the compensation) — the old bundle is exactly affordable in the new situation. *Second term is positive:* x is affordable at (p', w') while x' is chosen there, so by WARP x' was *not* affordable at (p, w) : $p \cdot x' > w = p \cdot x$, i.e. $p \cdot (x' - x) > 0$. Together: $(p' - p) \cdot (x' - x) = 0 - (\text{positive}) < 0$. ■

(*Law $\Rightarrow$ WARP*) is tedious and not extremely insightful; see MWG (as in the lecture).

R4 Proposition (MWG 2.F.2). WARP $\Rightarrow$ S negative semidefinite

Statement. If x is h.d.0, satisfies Walras' law and WARP, and is differentiable, then $v \cdot S(p, w) v \leq 0$ for all $v \in \mathbb{R}^L$.

Proof. Fix (p, w) , take any direction $v \in \mathbb{R}^L$ and a small price step $dp = \varepsilon v$, *compensated*: $dw = x(p, w) \cdot dp$. Differentiability gives, to first order,

$$dx = D_p x dp + D_w x dw = [D_p x + D_w x x^\top] dp = S(p, w) dp.$$

The change is a compensated price change, so the compensated law (R3, which used WARP + h.d.0 + Walras) applies:

$$dp \cdot dx \leq 0 \quad \implies \quad \varepsilon^2 (v \cdot S(p, w) v) \leq 0.$$

Since $\varepsilon^2 > 0$ and v was arbitrary, $v \cdot S v \leq 0$ for all v . ■

R5 Proposition (MWG 2.E.1–2.E.3, 2.F.3). Singularity: $S p = 0$ and $p^\top S = 0$

Statement. If x is h.d.0 and satisfies Walras' law (WARP not needed), then $S(p, w) p = 0$ and $p^\top S(p, w) = 0$.

Proof. Pure accounting. *h.d.0*: $x(\alpha p, \alpha w) = x(p, w)$ for all $\alpha > 0$; differentiate in α at $\alpha = 1$:

$$D_p x p + D_w x w = 0. \tag{E1}$$

Walras: $p \cdot x(p, w) = w$ identically; differentiate in w , then in p :

$$p^\top D_w x = 1, \quad p^\top D_p x + x^\top = 0. \quad (\text{E2}, \text{E3})$$

Right-multiply by p : $Sp = D_p x p + D_w x (x \cdot p) \stackrel{\text{Walras}}{=} D_p x p + D_w x w \stackrel{(\text{E1})}{=} 0$. Left-multiply by $p^\top$: $p^\top S = p^\top D_p x + (p^\top D_w x) x^\top \stackrel{(\text{E2})}{=} p^\top D_p x + x^\top \stackrel{(\text{E3})}{=} 0$. ■

Lecture 2 — the preference-based route

Maintained hypothesis: $\succsim$ on $X = \mathbb{R}_+^L$ is rational, continuous and locally nonsatiated (LNS), represented by a continuous u ; $p \gg 0$, $w > 0$; for the EMP, $u > u(0)$. Convexity/strict convexity added where stated.

R6 Theorem. Existence of a solution to the UMP

Statement. If u is continuous and $p \gg 0$, the UMP $\max_{x \geq 0} u(x)$ s.t. $p \cdot x \leq w$ has a solution.

Proof. The budget set $B_{p,w} = \{x \in \mathbb{R}_+^L : p \cdot x \leq w\}$ is *bounded*: for each ℓ , $x_\ell \leq w/p_\ell$ (using $p \gg 0$). It is *closed*: an intersection of closed half-spaces with the closed orthant. Closed and bounded in $\mathbb{R}^L$ is compact, and $B_{p,w} \neq \emptyset$ since $0 \in B_{p,w}$. A continuous function attains a maximum on a nonempty compact set (Weierstrass). ■

Where it can fail: some $p_\ell = 0$ (unbounded budget set), or u discontinuous — lexicographic preferences again.

R7 Theorem (MWG 3.D.2). Properties of Walrasian demand

Statement. Under the maintained hypothesis, $x(p, w)$ satisfies: (i) *h.d.0*: $x(\alpha p, \alpha w) = x(p, w)$, $\alpha > 0$; (ii) *Walras' law*: $p \cdot x = w$ for all $x \in x(p, w)$; (iii) convex $\succsim \Rightarrow x(p, w)$ convex; strictly convex $\succsim \Rightarrow x(p, w)$ single-valued.

Proof. (i) $B_{\alpha p, \alpha w} = \{x : \alpha p \cdot x \leq \alpha w\} = B_{p,w}$: same constraint set, same objective, hence the same maximizers. (ii) Suppose $x \in x(p, w)$ with $p \cdot x < w$. Then a small ball around x lies inside the budget set, and LNS supplies y in that ball with $y \succ x$: an affordable strict improvement, contradicting optimality. So $p \cdot x = w$. (iii) Let $x, x' \in x(p, w)$, both with utility u^* . The mixture $x'' = \alpha x + (1 - \alpha)x'$ is affordable ($B_{p,w}$ convex) and quasiconcavity gives $u(x'') \geq u^*$, so $x'' \in x(p, w)$: the solution set is convex. If u is *strictly* quasiconcave and $x \neq x'$, then $u(x'') > u^*$, contradicting maximality of u^* ; so the solution is unique. ■

In L1 (i) and (ii) were assumptions about data; here they are consequences.

R8 Theorem (MWG 3.E.1). UMP $\iff$ EMP

Statement. (i) If x^* is optimal in the UMP at wealth $w > 0$, then x^* is optimal in the EMP at required utility $u(x^*)$, and the minimized expenditure is exactly w . (ii) If x^* is optimal in the EMP at $u > u(0)$, then x^* is optimal in the UMP at wealth $p \cdot x^*$, and the maximized utility is exactly u .

Proof. (i) By Walras' law, $p \cdot x^* = w$. Suppose x^* does not solve the EMP at $u(x^*)$: some x' has $u(x') \geq u(x^*)$ and $p \cdot x' < w$. By LNS there is x'' near x' with $u(x'') > u(x') \geq u(x^*)$ and still $p \cdot x'' < w$ — an affordable bundle beating x^* in the UMP. Contradiction. The minimized expenditure is $p \cdot x^* = w$. (ii) Since $u > u(0)$, $x^* \neq 0$, so $w' := p \cdot x^* > 0$. Suppose x^* does not solve the UMP at w' : some x' has $p \cdot x' \leq w'$ and $u(x') > u(x^*) \geq u$. By continuity, $\alpha x'$ with $\alpha < 1$ close to 1 still has $u(\alpha x') > u$, at cost $\alpha p \cdot x' < w'$ — cheaper and feasible in the EMP, contradicting minimality. Finally $u(x^*) = u$ exactly: if $u(x^*) > u$, the same scaling applied to x^* itself yields a cheaper feasible bundle. ■

Hence the value functions are inverses in their last argument: $e(p, v(p, w)) \equiv w$ and $v(p, e(p, u)) \equiv u$; and $h(p, u) \equiv x(p, e(p, u))$, $x(p, w) \equiv h(p, v(p, w))$.

R9 Theorem (MWG 3.E.2). Properties of the expenditure function

Statement. $e(p, u)$ is (i) homogeneous of degree one in p ; (ii) strictly increasing in u and nondecreasing in each p_ℓ ; (iii) concave in p ; (iv) continuous in p for $p \gg 0$.

Proof. (i) The constraint set $\{x : u(x) \geq u\}$ does not involve p ; scaling $p \rightarrow \alpha p$ scales the objective, hence the minimized value, by α . (ii) Suppose $u' > u$ but $e(p, u') \leq e(p, u)$, and let x' attain u' at cost $e(p, u')$. By continuity, $\alpha x'$ with $\alpha < 1$ near 1 still gives utility $> u$, at cost $\alpha e(p, u') < e(p, u)$: contradicts the definition of $e(p, u)$. (Nondecreasing in p_ℓ : raising p_ℓ weakly raises the cost of every feasible bundle.) (iii) Fix $\bar{u}$; let $p'' = \alpha p + (1 - \alpha)p'$ and let x'' be optimal at p'' . Then x'' attains $\bar{u}$, hence is *feasible* (not necessarily optimal) at prices p and at prices p' :

$$e(p'', \bar{u}) = p'' \cdot x'' = \alpha p \cdot x'' + (1 - \alpha)p' \cdot x'' \geq \alpha e(p, \bar{u}) + (1 - \alpha)e(p', \bar{u}).$$

(iv) Maximum theorem; see MWG. ■

Concavity is the substitution property in value-function form; it becomes $D_p h$ NSD at R14.

R10 Theorem (MWG 3.E.3). Properties of Hicksian demand

Statement. For $p \gg 0$: (i) $h(\alpha p, u) = h(p, u)$ (h.d.0 in p); (ii) no excess utility: $u(x) = u$ for any $x \in h(p, u)$; (iii) convex $\succsim \Rightarrow h(p, u)$ convex; strictly convex $\succsim \Rightarrow$ single-valued.

Proof. (i) Minimizing $\alpha p \cdot x$ and minimizing $p \cdot x$ over the same constraint set $\{x : u(x) \geq u\}$ have the same solutions. (ii) Suppose $x \in h(p, u)$ with $u(x) > u$. Since $u > u(0)$, $x \neq 0$, so $p \cdot x > 0$. By continuity, αx with $\alpha < 1$ near 1 still has $u(\alpha x) > u$ — feasible and strictly cheaper. Contradiction. (iii) The constraint set is convex when u is quasiconcave and the objective is linear: two minimizers cost the same, and any mixture of them is feasible and costs the same, so the solution set is convex. Under *strict* convexity, the mixture of two distinct minimizers lies strictly inside the upper contour set, so a small radial contraction stays feasible and cuts cost — forcing uniqueness. ■

R11 Theorem (MWG 3.E.4). The compensated law of demand, from preferences

Statement. With $h(p, u)$ single-valued for $p \gg 0$: for all p' and p'' , $(p'' - p') \cdot [h(p'', u) - h(p', u)] \leq 0$.

Proof. Fix u . Each of $h(p', u)$ and $h(p'', u)$ attains utility u , so each is *feasible* in the other's problem.

Cost-minimality therefore gives

$$p'' \cdot h(p'', u) \leq p'' \cdot h(p', u), \quad p' \cdot h(p', u) \leq p' \cdot h(p'', u).$$

Add the first to minus the second: $(p'' - p') \cdot [h(p'', u) - h(p', u)] \leq 0$. ■

Two lines of feasibility — no calculus, no WARP. Same law as R3, second derivation, deeper hypothesis; they collide at R15.

R12 Proposition. Konüs $\leq$ Laspeyres

Statement. With $x^0 = x(p^0, w)$ and $u^0 = v(p^0, w)$:

$$P_K(p^0 \rightarrow p^1; u^0) = \frac{e(p^1, u^0)}{e(p^0, u^0)} \leq \frac{p^1 \cdot x^0}{p^0 \cdot x^0} = P_L.$$

Proof. x^0 attains u^0 , so it is feasible in the EMP at (p^1, u^0) : $e(p^1, u^0) \leq p^1 \cdot x^0$. And $e(p^0, u^0) = p^0 \cdot x^0 = w$ by R8(i) and Walras' law. Divide. ■

R2 in the answer's own language: fixed-basket (Laspeyres-type) indices mechanically over-state the rise in the cost of living.

Lecture 3 — the bridges

Add: $h(\cdot, u)$ single-valued, interior and differentiable where derivatives are taken; u differentiable where needed.

R13 Theorem (MWG 3.G.1). Shephard's lemma

Statement. For all $p \gg 0$ and u : $h(p, u) = \nabla_p e(p, u)$.

Proof. Write $e(p, u) = p \cdot h(p, u)$ and differentiate in p :

$$\nabla_p e(p, u) = h(p, u) + [D_p h(p, u)]^\top p.$$

The second term vanishes: the EMP first-order conditions give $p = \lambda \nabla u(h(p, u))$ for some $\lambda \geq 0$; “no excess utility” (R10(ii)) holds identically in p , so differentiating $u(h(p, u)) = u$ in p gives $[D_p h]^\top \nabla u(h) = 0$; hence $[D_p h]^\top p = \lambda [D_p h]^\top \nabla u(h) = 0$. ■

Envelope logic made exact: re-optimization is orthogonal to the price vector, so only the direct effect — the bundle itself — remains.

R14 Theorem (MWG 3.G.2). Properties of $D_p h$

Statement. With $h(\cdot, u)$ continuously differentiable at (p, u) : (i) $D_p h(p, u) = D_p^2 e(p, u)$; (ii) $D_p h(p, u)$ is symmetric and negative semidefinite; (iii) $D_p h(p, u) p = 0$.

Proof. (i) Shephard (R13): $h(p, u) = \nabla_p e(p, u)$ identically; differentiate in p . (ii) $e(\cdot, u)$ is twice continuously differentiable here, so its Hessian is symmetric (Young's theorem); and e is concave in p

(R9(iii)), so its Hessian is negative semidefinite. Both transfer to $D_p h$ by (i). (iii) $h(\cdot, u)$ is h.d.0 in p (R10(i)). Differentiate $h(\alpha p, u) = h(p, u)$ in α at $\alpha = 1$ (Euler): $D_p h(p, u)p = 0$. ■

R15 Theorem (MWG 3.G.3). The Slutsky equation

Statement. For all (p, w) and $u = v(p, w)$:

$$\frac{\partial h_\ell(p, u)}{\partial p_k} = \frac{\partial x_\ell(p, w)}{\partial p_k} + \frac{\partial x_\ell(p, w)}{\partial w} x_k(p, w), \quad \text{i.e.} \quad D_p h(p, u) = S(p, w).$$

Proof. Duality (R8) gives the identity in p : $h_\ell(p, u) = x_\ell(p, e(p, u))$. Differentiate both sides with respect to p_k (chain rule on the second argument):

$$\frac{\partial h_\ell(p, u)}{\partial p_k} = \frac{\partial x_\ell(p, e(p, u))}{\partial p_k} + \frac{\partial x_\ell(p, e(p, u))}{\partial w} \cdot \frac{\partial e(p, u)}{\partial p_k}.$$

By Shephard, $\partial e / \partial p_k = h_k(p, u)$; and at $u = v(p, w)$ we have $e(p, u) = w$ and $h_k(p, u) = x_k(p, w)$. Substitute. ■

The meeting point: the choice-built S of R4–R5 is the preference-built $D_p h$ of R14. Preference maximization therefore implies S symmetric, NSD and $S p = 0$ — symmetry being the one restriction beyond WARP's implications: the demand-level fingerprint of transitivity.

R16 Theorem (MWG 3.G.4). Roy's identity

Statement. If v is differentiable at $(\bar{p}, \bar{w}) \gg 0$ with $\partial v / \partial w > 0$, then for every ℓ :

$$x_\ell(\bar{p}, \bar{w}) = - \frac{\partial v(\bar{p}, \bar{w}) / \partial p_\ell}{\partial v(\bar{p}, \bar{w}) / \partial w}.$$

Proof. Let $\bar{u} = v(\bar{p}, \bar{w})$. Duality gives the identity in p : $v(p, e(p, \bar{u})) = \bar{u}$. Differentiate with respect to p_ℓ at $p = \bar{p}$ (chain rule):

$$\frac{\partial v(\bar{p}, \bar{w})}{\partial p_\ell} + \frac{\partial v(\bar{p}, \bar{w})}{\partial w} \cdot \frac{\partial e(\bar{p}, \bar{u})}{\partial p_\ell} = 0.$$

By Shephard, $\partial e / \partial p_\ell = h_\ell(\bar{p}, \bar{u}) = x_\ell(\bar{p}, \bar{w})$ (duality at $\bar{u} = v(\bar{p}, \bar{w})$). Divide by $\partial v / \partial w > 0$ and rearrange. ■

Read forwards: demand from v by differentiation. Read backwards: an ODE recovering v from an estimated demand — Hausman's engine (L4).

R17 Theorem. No wealth effects $\Rightarrow CV = \Delta CS = EV$

Statement. Suppose only p_1 changes, from p_1^0 to p_1^1 , and $\partial x_1(p, w) / \partial w = 0$ on the relevant range. Then $CV = \Delta CS = EV$, where ΔCS is the change in Marshallian consumer surplus.

Proof. Use $h_1(p, u) = x_1(p, e(p, u))$ and differentiate in u :

$$\frac{\partial h_1(p, u)}{\partial u} = \frac{\partial x_1}{\partial w} \cdot \frac{\partial e(p, u)}{\partial u} = 0.$$

So h_1 does not depend on u : $h_1(p, u^0) = h_1(p, u^1) = x_1(p, w)$ for every p on the path. The three areas are integrals of the same function:

$$-CV = \int_{p_1^0}^{p_1^1} h_1(t, u^0) dt = \int_{p_1^0}^{p_1^1} x_1(t, w) dt = \int_{p_1^0}^{p_1^1} h_1(t, u^1) dt = -EV. \quad \blacksquare$$

The license for Marshallian surplus areas — plausible for small budget shares, which is an empirical statement about *who you are*.

Lecture 4 — recovery (proof sketches)

R18 Theorem (integrability; MWG 3.H). Demand with the right properties is rationalizable

Statement. If $x(p, w)$ satisfies Walras' law, h.d.0, and has a symmetric, negative semidefinite Slutsky matrix, then there exists a rational preference relation rationalizing x .

Proof sketch. *Step 1 (recover e from x).* Fix (p^0, w^0) and solve the system of PDEs given by Shephard run backwards:

$$\nabla_p e(p) = x(p, e(p)), \quad e(p^0) = w^0.$$

A solution exists locally iff the system is consistent — Frobenius' condition: the cross-derivatives of the right-hand side must commute, which computes to exactly *symmetry of S* . Symmetry is the solvability condition for reconstructing the cost of living from demand. *Step 2 (recover preferences from e).* Given $e(\cdot, u)$ — concave in p by NSD — define upper contour sets as intersections of half-spaces:

$$V_u = \{x \in \mathbb{R}_+^L : p \cdot x \geq e(p, u) \text{ for all } p \gg 0\}.$$

Concave-duality (support-function) arguments show the preference relation these sets induce is rational and generates x . $\square$

Full argument: MWG 3.H.1–3.H.2.

R19 Theorem (Afriat 1967). Finite data: GARP $\iff$ concave rationalizability

Statement. For a finite dataset $\{(p^t, x^t)\}_{t=1}^T$, the following are equivalent: (i) the data satisfy GARP; (ii) there exist U_t and $\lambda_t > 0$ with $U_s \leq U_t + \lambda_t p^t \cdot (x^s - x^t)$ for all s, t (the Afriat inequalities); (iii) the data are rationalizable by a continuous, increasing, concave utility function.

Proof sketch. *(iii) $\Rightarrow$ (i):* a rationalizing utility orders the data; revealed-preference chains inherit the order, so no cycle can end strictly above its start. *(i) $\Rightarrow$ (ii):* GARP makes the revealed relation acyclic, so observations can be ordered; a combinatorial induction then picks $U_t, \lambda_t > 0$ satisfying the inequalities — read as “a concave function with slope $\lambda_t p^t$ at x^t leaves room for U_s at x^s .” *(ii) $\Rightarrow$ (iii):* build the utility explicitly as the lower envelope of the supporting hyperplanes:

$$u(x) = \min_t \{ U_t + \lambda_t p^t \cdot (x - x^t) \}$$

— continuous, increasing, concave by construction, and it rationalizes every observation. $\square$

Afriat (1967); Varian (1982). The CCEI relaxes (i): the largest budget scaling $e^* \leq 1$ at which GARP holds.

Stated in the block, not proved

- **Debreu's theorem** (rational + continuous $\Rightarrow$ continuous utility representation): MWG 3.C.1.
 - **Properties of indirect utility** (h.d.0; monotone; quasiconvex; continuous): MWG 3.D.3.
 - **MWG 2.F.1, direction (ii)** (compensated law $\Rightarrow$ WARP): MWG, Ch. 2.
 - **Houthakker's theorem** (SARP $\iff$ rationalizability for demand functions): Houthakker (1950); MWG 3.J. For $L = 2$, WARP $\iff$ SARP: MWG Ex. 3.J.1.
-